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RESEARCH ARTICLE

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Classical versus reinforcement learning algorithms for unmanned aerial vehicle network communication and coverage path planning: A systematic literature review

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Summary

The unmanned aerial vehicle network communication includes all points of interest during the coverage path planning. Coverage path planning in such networks is crucial for many applications, such as surveying, monitoring, and disaster management. Since the coverage path planning belongs to NP-hard issues, researchers in this domain are constantly looking for optimal solutions for this task. The speed, direction, altitude, environmental variations, and obstacles make coverage path planning more difficult. Researchers have proposed numerous algorithms regarding coverage path planning. In this study, we examined and discussed existing state-of-the-art coverage path planning algorithms. We divided the existing techniques into two core categories: Classical and reinforcement learning. The classical algorithms are further divided into subcategories due to the availability of considerable variations in this category. For each algorithm in both types, we examined the issues of mobility, altitude, and characteristics of known and unknown environments. We also discuss the optimality of different algorithms. At the end of each section, we discuss the existing research gaps and provide future insights to overcome those gaps.

KEYWORDS

air 2 ground, coverage path planning, network communication, reinforcement learning, unmanned aerial vehicles

1 | INTRODUCTION

It is challenging to manage the disaster relief operations like search and rescue, forest fires, tsunamis, and earthquakes across the globe. These operations have unique challenges and constraints in terms of time and resources, making them complicated to handle. The environment also plays an adversary role during relief operations. In these operations,

Abbreviations: A2G, air 2 ground; CPP, coverage path planning; RL, reinforcement learning; UAV, unmanned aerial vehicles.

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operational time efficiency is vital to save human lives. Since residential and communication facilities have sustained significant damage, human volunteers face two challenges in these cases. First, it is impossible to cover the entire area affected by a natural catastrophe physically, and second, communication among volunteers in the afflicted area is poor.

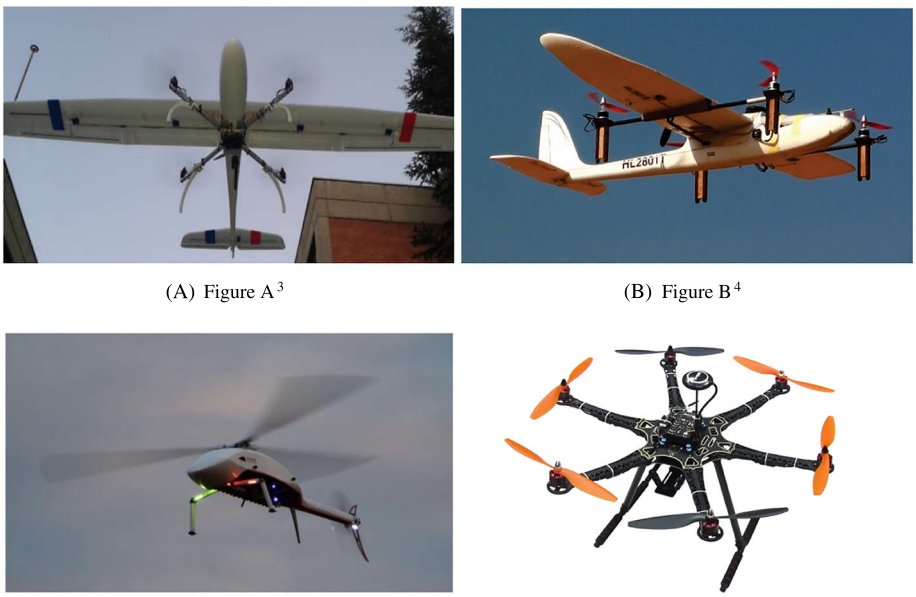
Coverage of the entire area is especially critical in emergencies. Human lives are valuable, and victims' access should be rapid and efficient. Therefore, all sites of interest in a given region or environment must be approached during coverage path planning. Graph theory defines it as visiting all the nodes in a particular graph. In these scenarios, unmanned aerial vehicles (UAVs) are successfully deployed to cover the whole site quickly and to erect the ad hoc communication infrastructure for the field workers and ground users.[1] UAV is a pilot-less aircraft. The UAV has different sizes as shown in Figure 1. The main components of a UAV are aircraft, sensors in the form of a camera or some other data collection equipment, a power battery, a ground station, and some other payloads.[2]

Figure 1 shows major types of drones also called UAV. Now a days, most commonly used UAV is multirotor UAV.

The success of UAVs in emergency operations is due to their fast and agile nature. The utilization of UAVs in firefighting and military is prevalent. In most applications, UAVs are continuously controlled from land and do not carry any human beings. UAVs effectively explore the target areas with the assistance of sensors like LiDARs, GPS systems, ultrasonic sensors, and cameras. This equipment is essential for the accurate measurement of location for data gathering. But continuous communication with the ground station creates further issues for the efficient deployment of UAVs. For example, UAV signals have limited range. If the distance between the UAV and the ground station is not managed, a UAV will be unable to decide its path and data gathering. The interaction among UAV, Environment and ground user is elaborated in Figure 2.

Figure 2 shows how the UAV is deployed and, in the target area, how the ground users have different positions in the environment.

Known and unknown environments have their own implications. For known environments, we already have information about the distribution of human population, type of terrain, and so forth. For known environments, normally informed search-based algorithms are deployed. For unknown environments, sampling-based algorithms are more suitable as we do not have any information which could be used to search the available space. Furthermore, UAVs are in perpetual motion during their flight to cover a maximum area, so it's unrealistic to invest a big chunk of resources to compute the distance among nodes. Instead, they continuously calculate the distance between the UAV and the ground station during the UAVs battery life and ensure maximum area coverage taxing power resources. Similarly, environmental distortions hinder the smooth communication between UAVs and ground stations. These issues motivate researchers to find a solution that can make a UAV autonomous during flight decision-making. Researchers have devised many efficient algorithms to make optimal path decisions for the flight duration, with the computational cost being the trade-off. This high cost is not feasible for UAVs as they have minimal battery power and flying time. In these



(A) Figure A[3] (B) Figure B[4]

(C) Figure C[5] (D) Figure D[6]

FIGURE 1 Four major types of unmanned aerial vehicles

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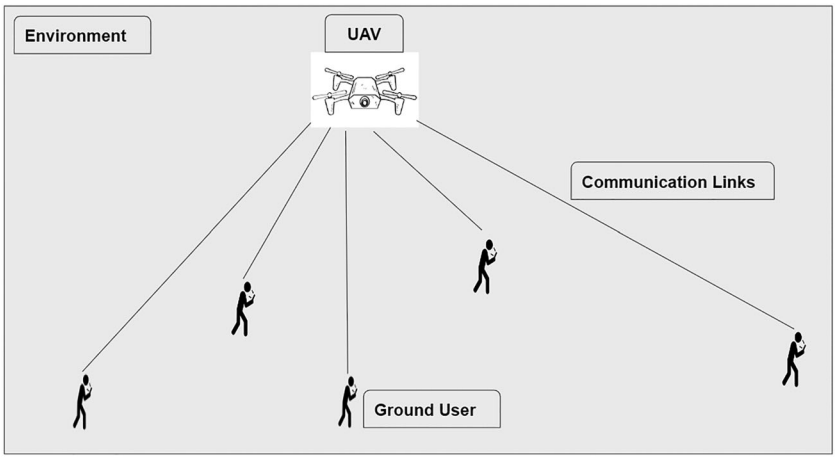

FIGURE 2 Representing the interaction among UAV, environment, and ground user

scenarios, RL surfaces as a viable option. RL is an excellent option to make UAVs autonomous. Like any other machine learning (ML) model, the RL model can make optimal decisions with optimal computation costs and resources. Making the UAV self-sufficient is the primary objective of RL. In ideal cases, UAVs are supposed to explore the region of interest autonomously by covering the whole area with limited resources. In ML, we assume that the agent, the UAV here, will learn from experience and improve its performance with time. It is emphasized that an autonomous UAV should learn from collected data from sensors and backgrounds that are based on trial and error. We can say that path or trajectory planning in the cases of static environments is a well-defined problem,[3,4] but for dynamic environments, there is still a need for in-depth investigations. Along with UAV path planning, the positioning techniques to locate the nodes that are reached by the UAV are very important. The positioning system proposed by the researchers in Kim et al[5] could improve the coverage path planning efficiency in terms of distance and energy efficiency in indoor environments. Still, many factors affect, influence, and control UAV path planning, where an on-ground human controller may have to intervene in path planning. Such interventions are further questionable in terms of optimization and time efficiency. The time needed for sending a request to the ground controller and getting the response causes further delays in path planning. Since the UAVs are designed for perennial unhindered motion during their battery lifetime, ground interventions and online path planning do not help.[6] Since the path planning and order of visits to different parts of a region belongs to np-hard problems in terms of nature, then optimizing path planning requires investigating an immense and conceivably boundless space of arrangements.

So far, it is evident that UAV coverage path planning becomes more challenging in case of following parameters,

1. Distance traveled by the UAV to cover the whole area
2. Energy consumption
3. Static and dynamic environments
4. 2D or 3D mobility
5. Obstacle avoidance
6. Automation

The study's primary goal is to offer in-depth knowledge of UAV coverage path planning algorithms. Since UAVs provide speedy and low-cost access to aerial data across a large area, responders can then map the search area and indicate possible hiding places for the missing person. Detect and recognize: Drones are outfitted with various sensors that can be used to recall objects on the ground. UAVs are employed for surveillance and tactical planning. In addition, this technology is now accessible to aid emergency response crew members. UAVs are categorized according to their altitude range, endurance, and weight and can be used for various military and commercial applications. In this study, we selected two main categories of coverage path planning algorithms for UAVs: Classical and RL-based algorithms. We chose these two categories because classical algorithms are optimal for UAV coverage path planning but cannot deal with dynamic parameters. Moreover, classical algorithms do not provide automation. On the other hand, RL algorithms are suitable for autonomous decision-making, but lack optimality and computational efficiency. This study investigates classical and RL based on coverage path planning algorithms. With these two algorithms, we analyze the limitations of

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both types of techniques concerning 2D and 3D spaces, performance efficiency in known and unknown terrains, obstacle avoidance, and working behavior in hostile environments. The performance comparison of both the categories as mentioned earlier of algorithms in the face of the mentioned hindrances paves the path to providing insightful results. We use the distance covered and time as the performance metrics for the performance analysis. Finally, we discuss in detail the learning results of the comparative analysis and the limitations of the coverage path planning algorithms. This study aims to provide knowledgeable insights into the topic and discuss the strengths and limitations of classical and RL algorithms.

2 | PLANNING PROTOCOL FOR COMPARATIVE REVIEW

We follow the method suggested by Templier[7] for this review study. The core sections of the method are given as follows.

1. Problem formulation
2. Searching the literature
3. Screening for inclusion
4. Assessing the quality
5. Data extraction
6. Results analysis

In problem formulation, we define the main objective of this study. In inclusion criteria, we specify the strings and keywords we used to search the existing literature. These keywords funnel the whole database, which results in the relevant literature. We ensure that we explore every possible repository to avoid missing any relevant articles. The following section is exclusion criteria, where out of all identified articles, we exclude some papers for logical reasons. In the quality parameters section, we define those rules on which we finalize a report as our core study paper. An article must possess some distinguishing characteristics to be included in the core study paper. We mainly focus on applied optimization techniques, dimensions, and known and unknown environments in data extraction fields. In the end, the execution and formulation of finalized search strings for use in search bars in literature repositories.

2.1 | Problem formulation

Based on the volume of the task, a single UAV or multiple UAVs are deployed. In both cases, a UAV communicates with ground users and ground control stations. For UAVs, path planning is critical to control mobility. In current UAV path planning models, the distance of the UAV from the ground control station and the ground user is not optimally managed. Continuous mobility of the UAV and ground user makes the path planning of the UAV complex to manage. If the mobility is not controlled, there is a long distance between UAVs, ground control stations, and ground users. Due to long distances, signal strength gets weaker, data packets are lost during communication, and UAV path planning models face the issue of high packet loss. Another problem in the UAV path planning model is the decision-making dependency of the UAV on ground control stations. All the executions are done on a ground control station, and for every move, the UAV gets instructions from the ground control station. The whole network is at risk if the communication between UAV and ground control station faces any problem. In this research, a UAV path planning model is proposed to deal with the mobility issues among all the devices in the UAV path planning model. This work aims to look into the shortcomings of RL and classical algorithms for UAV coverage path planning. Even though both types of algorithms are commonly used, they both have various inefficiencies. The comparison's performance measures include;

1. Distance traveled by the UAV to cover the whole area
2. Energy consumption
3. Static and dynamic environments
4. 2D or 3D mobility
5. Obstacle avoidance
6. Automation

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2.2 | Searching the literature

We explored authentic and massive databases related to computer science. We individually searched databases with specific strings and notations and then applied filters to sort out the relevant studies. The databases we looked into are included in Table 1.

Databases mentioned in Table 1 are the most followed sources in the computer science community. All of these publishing centers put out journals that are reviewed by peers and have high impact factors. All of the publications sent to these platforms go through a systematic review, and if they do not meet the standards, they are thrown out. We used Scopus and Web of Sciences as the validation tool for the papers we extracted from the above database sources. If any of the documents found in the above references do not exist on Scopus or WoS, that paper was not included in the review study. Instead, we applied our search strings in the following academic databases.

2.3 | Screening for inclusion and exclusion

At this point, evaluation of the selected papers is done. Inclusion and exclusion parameters are decided for the core studies on which the whole study will be built. The following are the terms used for inclusion.

2.3.1 | Inclusion criteria

Following are the core points to include a primary study.

1. CPP algorithms for UAVs based on classical techniques
2. CPP algorithms with RL
3. UAV coverage path planning for disaster management

2.3.2 | Exclusion criteria

Following are the core points to exclude a study.

1. Shortest path planning from point to point
2. Path planning in indoor environments
3. Papers not in the English language
4. Papers not listed in Web of Sciences or Scopus

TABLE 1 Search results of academic databases

Results of academic databases explored		
S. no.	**Database repository**	**Search portal**
1	IEEE	https://ieeexplore.ieee.org/Xplore/home.jsp
2	Wiley	https://onlinelibrary.wiley.com
3	ACM	https://dl.acm.org/
4	Springer	https://link.springer.com/
5	Elsevier	https://www.elsevier.com/search-results
6	MDPI	https://www.mdpi.com/
7	Sage	https://us.sagepub.com/en-us/nam/home

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2.4 | Assessing the quality

Here, we examine the quality of methodology opted by the existing research. If the methodology is strong enough, then the findings and results are reliable.

1. Is the paper relevant to UAV and coverage path planning?
2. Are the results clearly mentioned and compared with other related techniques?
3. Are the performance metrics clearly defined?
4. Is the research methodology clearly mentioned?
5. Is the article published in a reputed journal?

2.5 | Extracting data

At this point, we thoroughly study all the primary articles and extract those points which are important, common, and worthy of discussion.

1. What type of robots are being used in experiments?
2. Which algorithms were used for path planning?
3. Relevance with RL.
4. Single or multi drone deployment approach.
5. Short path planning or coverage path planning.
6. Dimensional involvement as 2D or 3D.

2.6 | Execution

The formulation of search strings is very important to conduct the effective literature review. Search strings are based on key words closely related to the topic of SLR. We used the following strings to find out the papers:

1. UAV algorithms for CPP
2. UAV dynamic path planning algorithms for covering the whole area
3. Path planning with RL
4. UAV trajectory planning with actor critic methods

2.7 | Results analysis

Results analysis is the last step according to our opted SLR methodology. However, this is the most essential and crucial part as well. We make analysis tables, describe the characteristics and limitations of existing methods, and suggest future directions for the researchers to contribute to this domain.

3 | LITERATURE REVIEW

Disaster management is not the only domain where coverage path planning has opted. There are many other real-world applications where coverage path planning techniques are deployed. At this point, we discuss the existing coverage path planning techniques in some major applications. Robots are used widely as vacuum cleaners.[8] During cleaning, a robot is supposed to cover the whole area required to be cleaned. A cleaning robot path is planned in such a way that, with minimum battery timings, the robot should visit all the cleanable points optimally. Atkar et al[9] discussed the techniques for robots used for painting vehicles. During painting, not even a small patch should be left without paint. CPP ensures that all the areas that should be painted must be covered. Window cleaning with robots is also in widespread use.[10] Underwater imaging with the help of robots is another new domain for the applications of CPP. In underwater imaging,

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initially, an area of interest is defined. Then the robot covers all the points of the area of interest and takes sample images.[11] In the case of an optimal CPP algorithm, there will be less time consumed to recharge the robot batteries repeatedly. Since UAVs are widely used in wireless communication networks, Lodhi et al[12] have defined efficient concurrent data communication strategies. An undersea and underwater survey is also as important as all other deployments of the UAV above the sea.[13] Lawn mowing is done quickly in case of proper path planning. On a lawn mower, a computational unit is attached, the CPP algorithm is implemented on this unit, and the lawn is mowed in a timely manner with minimum energy consumption.[14,15] Another similar application is crop harvesting and sowing.[16,17] A large area is sowed or harvested using robots. No point is left empty with the usage of minimum resources. Another major application of robotics is in the armed forces. Robots are used for neutralizing mines and surveillance.[18–20]

4 | DEFINITION OF KEY AREAS

At this step, we define the concepts of core topics, for example, UAV, coverage path planning, and the constraints that UAVs face in different applications. We also justify the need for this review paper. The main goal of this article is to look into the current state of the literature on coverage path planning in different real-world scenarios. We cover both static and dynamic UAV path planning. Static path planning has already been the subject of extensive research, leading to innovative shortest path algorithms such as the A-star and Dijkstra algorithms. These methods are also called classical techniques for path planning. Recently path planning research has shifted focus toward dynamic path planning. In dynamic path planning, the decision about the following waypoints for the path is calculated on the fly. In dynamic path planning, a path could be calculated with or without the coordination of the ground control station. Path planning without the control of the ground control station is more robust and automated as compared to the other case. In the case of the ground control station, the dependency on the ground control station itself creates new issues of linking and execution as the UAV is building up the latency in communication with the ground station and incurring colossal power consumption. Dynamic and autonomous path planning means the UAV makes decisions about the path, data gathering, and communication after starting from a particular point. For self-path planning, the UAV is equipped with some sensors like distance measures, GPS, cameras, and computational capability; the UAV decides its every next step to move forward. We shall be focusing on CPP algorithms. In CPP, the objective of a UAV is to touch every single cell in a particular area. A plethora of literature exists on the anonymity of CPP, yet human chaperoning is still required. The supervisor will be a human to control the take-off and landing of the UAV due to flight safety measures. CPP still has enormous challenges to tackle, like the burden of immense state-space exploration for optimal paths and covering the whole area with minimum repetitions to save the battery timings. For our SLR, we include papers from 2016 to the present, as 2016 was a milestone year in the history of RL. The match between Lee Sedol and AlphaGo marks a turning point in the development of artificial intelligence.[21] Table 2 defines the notations which are used in this paper.

4.1 | Analysis of the environment representation

The terrain or landscape of the target area is critical. There will be fundamental changes in the algorithmic approach if the terrain is different from the algorithm's design. For example, suppose an algorithm is designed to fly over a plane; in that case, it's not feasible and optimal for mountainous terrain as there will be communication and linking issues due to variations in the locality of ground nodes.

TABLE 2 Notations

Notations	
Notation	**Explanation**
$\mathbb{R}^3$	Represents all possible coordinate points in 3D
P	Represents the occupancy grid of the environment
Z	Represents obstacle coordinates
$z_e(T)$	Represents altitude of the UAV
x and y	Position coordinates of the UAV

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Figures 3 and 4 show the plain and nonplain terrains.

The environment where a drone is deployed for path planning is represented as a grid. This grid is decomposed into square cells. Typically, all the square cells have the same size, but it's not a rule, and the size of grids might vary in case of nonplain terrain. Our approach is based on a heuristic search where we know the environment is based on historical data, so for the sake of experiments, we prepared probability grids. Probability ranges from 0.1 to 0.99, with 0.99 representing the highest probability for the target existence and 0.1 showing the lowest chance. In the following sections, we sketch different probability grid maps with varying styles of probability distributions. We have a target area to cover in real-world scenarios in UAV operations. The whole area is represented in the form of grids. Grids are decomposed into cells. Each cell has unique numbers showing the probability of human existence value of signal strength.

4.1.1 | Impacts of decomposing the environment

The geography in the area could be flat, canyon-like, or steep. Breaking down more significant target sites into more manageable portions is one of the first tasks in route planning. The cell sizes are computed based on the camera's viewing angle and the aircraft's height to improve the UAVs performance. Another critical aspect of the environment is whether or not the various parts of the searchable area are fully connected or disconnected within a single search space. For example, if the searchable site has human population patches, the UAV is expected to cover all of the human

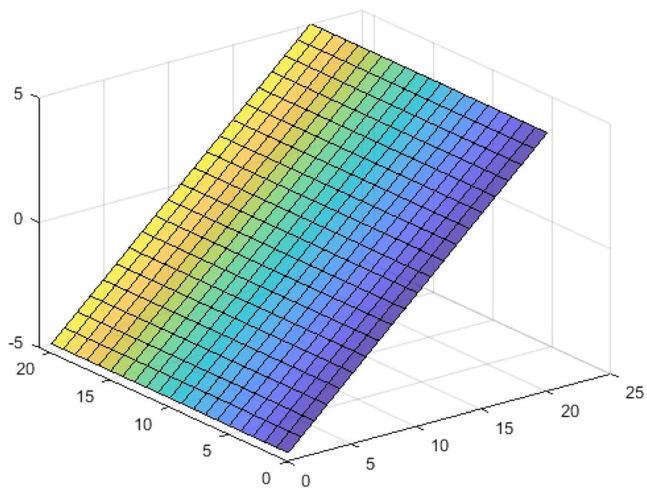

FIGURE 3 Representing the plain terrain in three-dimensional environment

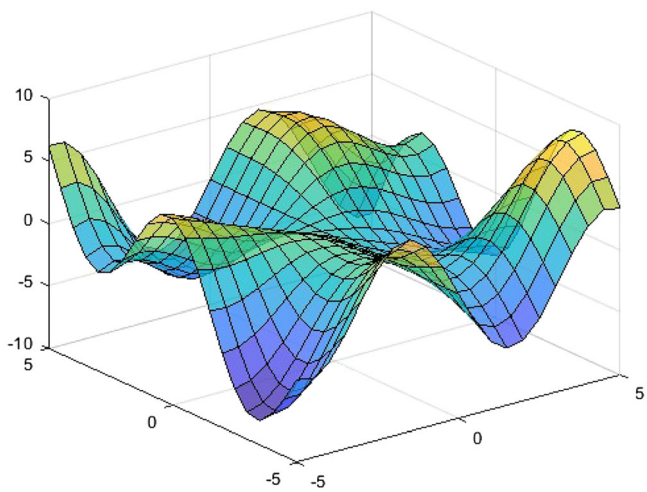

FIGURE 4 Representing the nonplain terrain in three-dimensional environment

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population patches; these population patches are regarded as separate parts. The particular area could also take the form of islands, populated islands with populations separated from one another. Another critical aspect of the environment is the availability of previous historical data from the searchable area. This historical record reveals which areas have a high probability of human existence. For the sake of experiments, we decompose the searchable space into smaller and manageable square cells. These cells form a grid world of size

$$RxR \in G^2,$$

where the size of each cell is s. The set G contains all possible positions. Therefore, the decomposition of the area is significant. The idea of decomposition is based on algorithmic theory, where a big task is divided into smaller parts to reduce the complexity and enhance efficiency. Decomposition of the environment makes it feasible for the UAV to plan a path step by step. The decomposed square cells are the steps here. The takeoff and landing point for the UAV is also essential and represented as P.

$$P = \{[x_e^n, y_e^n]^T, e = 1,..., N, : [x_i^n, y_i^n]^T \in G\}. \tag{1}$$

All those positions where the obstacles are expected, or in the case of urban environments, the positions of buildings are treated as no-fly-zones. These zones are represented as

$$Z = \{[x_e^m, y_e^m]^T, e = 1,..., N, : [x_i^m, y_i^m]^T \in G\}. \tag{2}$$

5 | DISTANCE BETWEEN TWO POINTS

$$h(x) = (end_x - P_x)^2 + (end_y - P_y)^2. \tag{3}$$

In Equation (3), the UAV calculates the distance from its current location to the remaining unvisited waypoints and chooses the shortest one.

5.1 | Commonly used drone model in path planning

Set represented by $\boldsymbol{A}$ ensures that all UAVs are mobile within the limits of grid world $\boldsymbol{G}$. At any given time $\boldsymbol{T}$, to map the eth position of the UAV in 3D, this equation is used

$$L_e(A) = [x_e(A), y_e(A), z_e(A)]^T \in \mathbb{R}^3, \tag{4}$$

here, $\boldsymbol{A}$ represents agent which is UAV, $\boldsymbol{L}$ denotes location as the 3rd dimension being taken as altitude. Altitude of the UAV from land is represented as

$$z_e(T) \in \{0, h\}.$$

For the third dimension, the UAV is always on ground or at some particular height, which is not changed during flight. The change in altitude depends on the algorithm for path planning whether it is two or three dimensional.

6 | CHARACTERISTICS OF COVERAGE PATH PLANNING FOR UAV

Any moving robot needs path planning. The same is the case with UAVs. If we want to access some specific point, the UAV should know its current position in the environment, called localization. Another factor is the environment; the environment is the area where the UAV is currently operating. After localization, the UAV should know its destination,

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also called the target location. In our case, there is no target point, but a path because the goal is to cover the whole area. Such problems fall under the scope of coverage path planning (CPP). Finding the shortest way from point A to point B is a process used when we are trying to locate a specific objective. Coverage is not the goal here. In the case of coverage path planning, coverage of the whole area with minimum resources is the goal.[2] In CPP, a UAV must cover all the points of interest within optimal time and with efficient resource utilization. CPP is critical for many applications, including surveying, monitoring, and disaster management operations. Coverage path planning belongs to NP-hard problems, so the researchers of this domain are always trying to find optimal algorithms for CPP.

6.1 | Analysis of classical coverage path planning algorithms

The following three categories are used to discuss the overall classical algorithms for clarity..

1. Graph-search-based algorithms
2. Sampling-based algorithms
3. General classical path planning algorithms
4. Bio-inspired algorithms

Table 3 compares the various graph search strategies in terms of path optimality, computing speed, real-time capability, memory use, safety, and completeness.

TABLE 3 Analysis table for graph-search-based classical path planning techniques

Reference	Algorithm	Application	Characteristics and limitations
Debnath et al.[22]	Dijkstra	Shortest path finding	Single UAV approach % Total area is divided into sub-areas for survey. % There is no specific target.
Mandloi et al.[23]	A*	UAV path planning and it is used in other algorithms such as the Bellman-Ford algorithm to solve the shortest path problem.	It is optimal search algorithm in terms of heuristics. It is one of the best heuristic search techniques. It is used to solve complex search problems. There is no other optimal algorithm guaranteed to expand fewer nodes than A*.
Qadir et al.[24]	D*	Mobile vehicles navigation. Optimal path in hierarchical structures. Shortest path finding	This algorithm has the ability to re-plan a path if the first path is blocked by using the algorithm's previous calculations. This increases the likelihood of successfully navigating the maze.
Tang et al.[25]	Depth first search	Topological sorting, scheduling problems, cycle detection in graphs, and solving puzzles with only one solution such as a maze or a sudoku puzzle.	Traversing or searching tree or graph data structures is simple. In the case of a graph, the algorithm starts at the root node by selecting any arbitrary node as the root node and explores as far as possible along each branch before backtracking.
Li et al.[26]	Breadth-first search	Finding the shortest path in a graph and solving puzzle games (such as Rubik's Cubes)	Best advantage is to solve the problem in an iterative way. When a dead end occurs during any iteration, BFS uses a queue to keep track of where to receive the next vertex to begin a search.
Wu et al[27] and Bal[28]	Best first search	Used for searching the best option in hierarchical way, in games for move optimization, web crawlers.	This algorithm explores a graph by expanding the most promising node based on a predefined rule.

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6.1.1 | Lessons learned from graph based classical path planning techniques

It is evident in Table 3 that graph search-based algorithms have been extensively used in previous energy efficiency studies. It checks some nodes/states to determine a path from starting to target points. It will report failure if there is no existing path.

Table 3 shows that the Dijkstra algorithm returns the shortest path between two nodes. It cannot estimate the distance between each node and the target as an optimal solution in all those cases where no prior knowledge of the graph is available. Dijkstra's algorithm typically covers a large area in the graph due to its selection of edges with the lowest cost at each step, which is vital in situations where multiple target nodes have no prior knowledge of the closest one. Dijkstra's search algorithm was introduced in 1959 to find an optimal path between the initial and all other points in the graph-based approach with the lowest traversal costs in terms of distance traveled.

Hart et al[29] put forth the heuristic search algorithm A* in 1968. The total cost cost function is represented as $f(n) = g(n) + h(n)$, calculated using the forward cost function $h(n)$ and the backward cost function $g(n)$ between the source and destination nodes in this algorithm. The A* combines the features of the Best-first search and Dijkstra's algorithm. To determine the heuristic value that is a straight line distance, it typically ignores the obstruction between the current and destination nodes.[30] When backward cost is dominant, and the shortest path between the source and target nodes is found, A* appears to be more similar to Dijkstra's algorithm. In extreme instances, A* transforms into Dijkstra's Algorithm for zero heuristic and longer-duration searches. A* is not very efficient because it must be run multiple times for each target node to obtain the optimal solution out of all possible cases. A* only gives more importance to those nodes which are promising in terms of distance and time cost. Its approach is similar to greedy, where the algorithm always pics the most optimal value available without considering the future implications. Other shortest path finding techniques could be proposed based on the assessment strategies of A*.

The D* method divides the search space into open and closed lists. Open means not visited, and closed means visited nodes. Iteratively, the algorithm selects and evaluates a node from the open list. The node's modifications are then propagated to all its neighbors, who are added to the open list. In D* algorithm, each node or waypoint is given a unique label by the label propagation algorithm (LPA), which updates each node's tags based on its neighbors until no tags are changed. The linear computational complexity of LPA is its most significant benefit, but its uncertainty and randomness are also substantial. Every node is initialized with a distinct community label in the LPA algorithm (an identifier). They spread throughout the network. After each propagation iteration, each node updates its tag to the one to which most of its neighbors belong. The artificial potential field algorithm is one of the algorithms used in UAV path planning approaches. Using the concept of attracted and repulsive force methods, the upgraded versions of this method are used to efficiently avoid dynamic impediments and attain the objective by utilizing attractive and repulsive force methods. In an unknown environment, attractive force to accomplish the goal, and repulsive force to avoid barriers.

Depth First Search (DFS) examines a path until it comes to a dead end while moving as swiftly as practicable in the direction of the goal. DFS tries to search multiple pathways at once before finishing one path. Therefore, it may overlook significant chunks of the workspace.[31,32] It can be used to select a path from many alternatives. However, because the cost function is not employed to determine the best direction or to calculate how far the target point is from the current node, DFS is an uninformed search. DFS is useful for selecting only one solution on a greedy basis from a large number of options without concerning the issue of optimality. But if only one solution or shortest path is required, this technique is not feasible. With DFS, a solution can be obtained by conducting the fitness of the most feasible solution across all nodes.

Moore developed the breadth-first search (BrFS) algorithm in 1957.[33] It is a systematic search algorithm since it enlarges the shallow nodes first by looking through all of the path's next-level nodes before moving on to the next step. BrFS is an uninformed search to locate the shortest path in the first attempt, much like DFS.[34] It is useful in limited solutions with a small number of steps. For those outcomes that are scarcely accessible and involve a manageably low number of steps, a BrFS is appropriate. BrFS is adequate for limited available solutions with a small number of steps. Its distinctive feature finds the shortest path from the source node to the destination. This technique also considers the graph's edges either unweighted or have similar weight. BrFS is beneficial because it avoids dead ends. Best-first search (BeFS) is a heuristic search algorithm that uses the distance from a current node to the target point to find a path in a graph. BeFS is a graph search algorithm that selects a node for expansion based on an evaluation function.[35] A heuristic is used to estimate such distance. Because the distance to the goal is computed, the node with the lowest evaluated cost is usually chosen to explain. Since it skips some branches in the search tree, the BeFS algorithm cannot guarantee that

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it will find the shortest path. It is a greedy search that is neither complete nor optimal. It is used due to its time efficiency. In Table 4, we show the sampling based path planning algorithms and their characteristics.

6.1.2 | Lessons learned from sampling-based classical path planning techniques

According to the results of Table 4, the rapidly exploring random tree (RRT) and its variations are frequently employed in UAV path planning due to its superior search performance in crowded situations. RRT and its variants are sampling-based techniques. RRT is commonly used in UAV pathfinding because it can discover a path faster, even in unexplored territory. Since the algorithm evaluates a large number of options to select the path, the path may not be ideal due to its wide search space and high randomness. RRT is applied to environments that are transformed into the grid. Grids are further decomposed into cells. Due to the huge sampling space, RRT faces trouble devising the optimal path from cell to cell while covering the entire region of interest. As a result, few advances in RRT have been made by simplifying the cell connection strategy to generate a feasible path that meets the operating environment. RRT* overcomes this nonoptimality deficiency.[37]

Compared to RRT, RRT* introduces the processes of reshaping and reselecting the parent node of the new node after the RRT algorithm identifies the new node. For collision avoidance and adequate area coverage, modified RRT is applied. As a result, potentially guided intelligent bidirectional RRT* (PIB-RRT*)[39] can reach the ideal outcome more quickly. Informed-RRT*[40] and RRT*-smart[41] choose higher quality random points during the sampling procedures in addition to enhancing the random tree's development mode to increase search efficiency. The optimal solution must be found when the number of iterations approaches infinity, but it has a high path cost and a slow convergence rate.[42,43]

TABLE 4 Analysis table for sampling-based classical path planning techniques

Reference	Algorithm	Application	Characteristics and limitations
Fan et al.[36]	Rapidly exploring random tree	Shortest path finding in unknown environments	% Rapidly exploring Random Tree is a sampling-based algorithm which is widely used in UAV path planning due to the excellent search performance in cluttered environments. Due to its large search space and strong randomness, the planned path is not optimal. % When the number of iterations approaches infinity, the optimal solution must be found, but it suffers from high path cost and slow convergence rate. When the number of iterations approaches infinity, the optimal solution must be found, but it suffers from high path cost and slow convergence rate.
Liao et al.[37]	Rapidly exploring random tree—star	Shortest path finding	Without requiring full knowledge of the obstacle space, it offers a probabilistically complete and asymptotically ideal solution. RRT* converges to the optimal solution rather slowly despite all of its benefits.
Huang and Sun[38]	Bidirectional rapidly exploring random tree	Optimal path finding in huge search space	Based on greedy strategy. Avoids obstacle. Provides quick response and formation reconstruction for UAV formations. Enhances the effectiveness and formation stability of the optimization technique in challenging conditions.
Tahir et al.[39]	Potentially guided intelligent bidirectional rapidly exploring random tree	Applicable to the highly constrained environments	To improve the convergence rate. Performs pure exploration. Tend to suffer in highly cluttered environments.
Meng et al.[40]	Informed rapidly exploring random tree	Shortest path finding	Informed RRT* is an improvement to the RRT* algorithm that increases the rate at which the found solution converges to the optimum. Developed specifically for problems seeking to minimize path-length.
Tak et al.[41]	Rapidly exploring random tree—smart	Shortest path finding	Provides asymptotic optimality. Slow convergence rate and costly initial solution makes it inefficient.

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Bidirectional RRT*[38] and intelligent bidirectional RRT* (IB-RRT*) are proposed to accelerate convergence. However, both are susceptible to interference from highly complex situations. This approach, which uses a greedy strategy, is suggested to implement obstacle avoidance, quick response, and formation reconstruction for UAV formations. It enhances the effectiveness and formation stability of the present optimization technique in challenging conditions. Its bidirectional counterparts, the bidirectional RRT* (B-RRT*) and intelligent bidirectional RRT* (IB-RRT*), were introduced to increase convergence rates. However, because both variations only engage in pure exploration, they often struggle in congested situations. Therefore, we propose potentially guided intelligent bidirectional RRT* (PIB-RRT*) and potentially guided bidirectional RRT* (PB-RRT*) to address these constraints by introducing a novel notion of potentially guided bidirectional trees. The proposed techniques significantly increase the convergence rate and better use of available memory.

6.1.3 | Lessons learned from general classical path planning techniques

In Table 5, it is evident that based on the state of the input, the fuzzy logic and adaptive neural fuzzy inference system provide the foundation to create an entire control system. The input's condition and pace of change impact the output. In other words, a fuzzy logic system determines the outcome based on the likelihood that the input state will occur. Support vector machines map data points to a high-dimensional feature space to categorize data points that are otherwise not linearly separable. The data are changed after the discovery of a separator between the categories such that the separator can be drawn as a hyperplane. The algorithm performs relatively well when there is an understandable margin of class dissociation. In high-dimensional spaces, it is more effective. When the quantity of dimensions exceeds the number of examples, it works well.

Another meta-heuristic method is the vortex search (VS) algorithm, which is based on the vertical flow of heated fluids. The VS technique does have significant drawbacks, even though it has been demonstrated to be a strong contender for tackling specific optimization issues. Model predictive control builds the model based on historical data and projected future input. This model is used to predict the future output. Instead of focusing on the model's structure, it highlights its functionality. The advantage of the MPC is that it is a multivariable controller that manages all of the interactions between the system variables and controls the outputs concurrently. Another strength of MPC is its ability to manage constraints. MPC can work effectively within the accurate actuators with relatively narrow conditions. The main disadvantage of MPC is its complexity, which requires more time than the other controller.

Adaptive cell decomposition is a powerful method for solving physical systems governed by partial differential equations. Such techniques reduce computational time significantly, and a dominant equations system does not control discretization citesamaniego2019recursive. The precise reconstruction of intricate 3D Cartesian geometry is accomplished with this technology. This method seeks to identify occupied and accessible places inside 3D Cartesian space rather than attempting a finely tuned environment reconstruction.[62] Planning dynamic paths based on a slicing-based adaptive algorithm is more effective at navigating the intricacies of a 3D world.[54] This method evenly tackles surface irregularities while designing the path. Irregularities mean sharp corners, holes, protrusions, concavities, and discontinuities.[63] It identifies and measures the significance of concavities and surface abnormalities and determines whether or not these should be taken into account in the path plan by contrasting the actual part geometry with the convex hull path.[55] With a dense population field, the spectra multi-scale coverage algorithm extracts the high-priority parts of the field and guides UAVs to those areas with tremendous potential. Implementing the distance field and adjusting the agent's directional vector when the distance threshold is crossed prevents collisions between the agents and the domain borders.[64] It is also possible to use this technique in 3D settings. Mixed integer linear programming (MILP) is a powerful mathematical modeling technique for handling challenging global optimization problems. The decision-makers can have a firm understanding of potential trade-offs between competing objectives, and MILP accurately identifies the trade-offs.[65]

6.2 | Analysis of bio-inspired classical coverage path planning algorithms

Recently, engineering research has begun to place a premium on the use of computational methods inspired by biology, in particular those that may be used to the resolution of difficult optimization issues. These methods of stochastic search are designed to get close to optimal results when solving big optimization puzzles. In Table 6, we discuss the important proposed techniques related to bio-inspired algorithms.

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TABLE 5 Analysis table for general algorithms for classical path planning techniques

Reference	Algorithm	Application	Characteristics and limitations
Pan et al[44] and Souza et al.[45]	Potential field	Robotics path planning in search and rescue operations	% A search and rescue based project % Probability based map is drawn % Four different Algo applied to visit high probability areas first. Potential field algorithm comparatively proved to be more efficient and smooth
Wang and Pan[46] and Selma et al.[47]	Fuzzy logic, ANFIS	UAV path planning	% A search and rescue based project % Probability based map is drawn % Four different Algo applied to visit high probability areas first. Potential field algorithm comparatively proved to be more efficient and smooth
Figetakis and Refaey[48] and Liu et al.[49]	Support vector machine with single shot multibox detector	Moving object tracking	Dynamic path planning and target tracking in real time. Deals with unknown environments. Optimization is done on the idea of elliptical tangent method. Algorithm performs well in the case of understandable margin of dissociation between classes. More productive in high-dimensional spaces.
Wang et al.[50]	Adaptive vortex search algorithm	Forest fire search and rescue	Specially designed for forest fire and search and rescue operations. Basic idea depends on Vortex Search Algorithm. Features of the terrain are extracted by cubic spline method. For efficient path planning, The Search of Optimal Path with Vortex Search is done along with a cost function which is multi objective.
Gräfe et al[51] and Yao and Zhao[52]	Model predictive control	Path planning in uncertain environments	Optimization is based on rolling feedback taking MPC technique as a base. Dynamic path planning. Sudden threats are dealt on the fly by computing alternative paths around the obstacle or threat. Other parameters for alternate paths are direction, speed and current position of the UAV.
Samaniego et al.[53]	Adaptive cell decomposition	General purpose	Environment is approximated by 3D ACD technique with structure as Octree. MADC technique is used for optimization. Obstacle are taken care by calculating alternate paths.
Brown and Pierson[54] and Mao et al.[55]	Slicing-based adaptive path planning	Industrial spraying operations	UGV-based autonomous path planning. The proposed technique especially focus on abnormalities of the environment in the form or sudden turns and sharp corners. A geometrically smooth path is designed having less number of turns to enhance time efficiency and to reduce energy consumption.
Ivić et al.[56]	Spectral multiscale coverage	Search strategy in a complex and dynamic environment	A multi-UAV approach. Optimization is based on Modified DSMC. Especially designed for dynamic and unpredictable environments. This model has the ability to deal with geometrical complexities and abnormalities in the terrains. Deals with sudden changes due to hostile environmental conditions.
Chen et al.[57]	Linear programming	Multiscenario	Heuristic Algorithm Suitable for heterogeneous UAVs. Linear Programming facilitates rational thought and offers deeper understanding of search space for the optimization problems.
Xiao and Wachs[58]	Maximum predictive error method	Multiscenario	Sampling based method, global path planning with the optimized nonmyopic exploration efficiency. Offline
Snyder et al.[59]	Sequential quadratic programming	Volcanic survey	Area decomposition into nonuniform importance regions. More important are visited first. Offline.
Apuroop et al[60] and Melo et al.[61]	Mixed integer linear programming (MILP)	Multiscenario	Proposed for bounded number of regions. For 3D environments, for unknown environments, energy efficient. Online.

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TABLE 6 Analysis of bio-inspired classical coverage path planning algorithms

Reference	Algorithm	Application	Characteristics and limitations
Israr et al.[66]	Particle swarm optimization	Shortest path finding	Bio-inspired techniques gives UAVs a path that takes into account three factors. These factors include the path reliability cost, journey distance, and cost incurred. The idea of swarm of UAVs is emerging rapidly. In swarm of UAVs, the collective behavior of all the agents is represented by the swarm. Despite having restricted powers, a swarm of agents can cooperate to complete tasks even when they are spread out at a distance. As a result, even in the face of uncertainties and complexity, quick, inexpensive, and ideal solutions can be found.
Zhu[67] and Rashid et al.[68]	Artificial immune system algorithm	Disaster management and target tracking	Efficient for binary classification. Provides better optimization performance.
Long et al.[69]	Bacteria foraging optimization algorithm	Deals with local minima and maxima problems	This improves search performance, especially in complex environments by dividing a problem into smaller sub problems. Deletion or immigration strategies are designed by ensuring that the optimized individual of each sub problem is inherited by others, resulting in a fast convergence speed.
Han et al[70] and Liu et al.[71]	Artificial bee colony algorithm	Search and rescue	Optimal Path Planning, The ABC algorithm, a swarm-based methodology, is an optimization technique inspired by honeybee foraging behavior. Honeybees are able to coordinate among themselves to achieve both a global and local optimum thanks to their self-organization ability.
Marcelino et al.[72]	Coral reef optimization algorithm	General purpose UAV path planning	Produces shortest path with optimal costs from source to goal state.
Hoang and Phung[73]	Teaching-learning-based optimization algorithm	Optimal path planning	The TLBO method does not require any algorithm parameters to operate, excluding the number of iterations and the size of the population.
Song et al[74] and Xie and Zheng[75]	Cuckoo search algorithm	Used for continuous non linear optimization	Cuckoo Search is a method with several advantages, including ease of use and fewer tuning parameters. Drawback is that it falls into local optimal solutions and has a slow rate of convergence.
Ali et al[76] and Shafiq et al.[77]	Ant colony optimization algorithm	To solve graph problems by investigating possible paths on the graphs	Inherent parallelism, Positive Feedback accounts for rapid discovery of good solutions, Efficient for Traveling Salesman Problem and similar problems. Disadvantage is that classical ant colony optimization performs slow convergence speed and the possibility of falling into a local optimum.
Cheng et al.[78]	Firefly algorithm	Firefly algorithm is applied for scheduling and traveling salesman problem, classifications, clustering, least computation time for digital	The Firefly algorithm is a simple and efficient algorithm. It can also be implemented in parallel. However, studies show that for

(Continues)

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TABLE 6 (Continued)

Reference	Algorithm	Application	Characteristics and limitations
		image compression, feature selection, antenna design optimization, multiobjective load dispatch problem	multimodal problems, it is slow to convergence and easily becomes trapped in the local optimum.
Pan et al.[79,80]	Shuffled frog leaping algorithm	Path optimization	A population-based algorithm that is used to solve complex and important optimization problems. The main idea behind this algorithm is to improve the aqueous performance of the search intensification process by using a local search method within the structure of the genetic algorithm.
Chen et al.[81]	Pigeon-inspired optimization algorithm	Applied to reduce the UAV energy consumption. Based on simple update principles iteratively, provides high flexibility, optimized way of search space evaluation	The advantages of this algorithm for optimization issues are that it is fast to process, doesn't require a lot of parameters, and is relatively simple.
Qu et al.[82]	Gray wolf optimization	UAV path planning, efficient space search	Improve the accuracy and convergence speed, and the optimization effect is better.

6.2.1 | Lessons learned from bio-inspired algorithms

The algorithms mentioned in Table 6 shows that natural patterns for path planning inspired by living things are an important area that scholars are working to understand. These are divided into categories based on the connection between biological evolution and other behavioral types. The main benefit of bio-inspired algorithms is that they can handle static and dynamic threats while maintaining offline functionality. Evolution-based algorithms and swarm-based algorithms are the two primary categories of bio-inspired techniques. UAVs are given a path using an evolution-based approach that considers three factors. These factors include the cost of path reliability, the length of the trip, and the expense incurred. In the initial generation, these evolving algorithms randomly select workable and doable solutions, and in the subsequent generations, they consider the parameters to explain whether the randomly selected workable solutions are appropriate or not. These algorithms include the Genetic Algorithm, Harmony Search Algorithm, Evolutionary Strategy, Evolutionary Programming, and Differential Evolution Algorithm. Based on three fundamental operations–selection, crossover, and mutation–the Genetic Algorithm develops the most effective and ideal path. In addition to these advantages, it may cause a protracted, premature convergence that yields less-than-ideal results. The genetic approach is not practical for dynamic path planning since it is not computationally optimal. Fogel created the EP method in 1990. For best outcomes, iterations are necessary. ES is another evolutionary algorithm that solves optimization issues by applying predetermined rules.[83]

The differential evolution algorithm uses actual coding rather than binary coding. It reduces the computational cost while improving the end path.[84] A search strategy called harmony search (HS) is based on how jazz musicians improvise.[85] In jazz, the various performers attempt to alter their pitches to optimize the overall harmonies for esthetic reasons.[86] Another category of bio-inspired algorithms is the swarm-based technique. These algorithms are based on the idea of two basic natural phenomena, a fundamental evolution of nature-based methods and population-based strategies.[87] First, the swarm is a representation of all agents' collective behavior. For example, a swarm of UAVs can work together to fulfill tasks even when they are separated by distance, despite having limited powers. As a result, quick, low-cost, and optimal solutions can be developed despite uncertainty and complexity. Commonly used bio-inspired techniques include ant colony optimization approach, bacteria foraging optimization technique, particle swarm optimization, artificial bee colony algorithm, shuffled frog leaping algorithm, artificial immune algorithm, cuckoo search technique, teaching learning optimization, pigeon-inspired optimization, coral reef optimization algorithm, and firefly algorithm.

AIS is a swarm-based intelligent algorithm based on the natural principles of the human immune system. It possesses immune system memory and learning abilities that it can employ to solve problems. It provides appropriate path planning trajectories with minimal processing. The concept of an insect crowd movement served as the foundation for PSO development. Every particle in the swarm searches for the optimal path, which is synchronized with the findings of the

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other swarm members and generates a velocity vector toward the goal point. This algorithm's main advantage is that it can develop optimal 3D path planning. However, premature convergence and excessive time complexity are drawbacks of this algorithm. The approach used by Passino[88] was based on how the *Escherichia coli* bacteria live in human intestines forage. He named this clever program BFO. It provides quick convergence as well as an international search. Xin-She Yang and Suash Deb[89] devised the cuckoo search (CS) optimization technique for the optimal space search in 2009. According to this notion, cuckoos lay their eggs in host birds of other species nests. Each egg in a nest of cuckoo eggs stands for a different approach to a problem, and this is the algorithm's fundamental model. The objective is to introduce new, maybe superior solutions instead of subpar ones in the nests. Each nest consists of one egg in its most basic form. With each nest having many eggs symbolizing a set of solutions, the algorithm can be expanded to handle increasingly complex instances. The ABC algorithm solves constraint-based optimization problems. This algorithm will take the shortest route to its destination while avoiding collisions with other drones. The algorithm can look for a different path to avoid a collision by setting a limit of the smallest permissible distance between UAVs.[70] An effective combinatorial algorithm for discrete applications is ACO. A new ant colony model was developed for continuous space; however, it is relatively complicated. The ACO algorithm is a meta-heuristic method for figuring out the shortest path while dealing with constant and multi-objective path planning challenges based on ant pheromones emitted by the ants during a food hunt. The CRO algorithm successfully resolves complex optimization problems and has several advantages.[90]

The coral reefs optimization (CRO) algorithm is a novel global-search method based on the biology of corals and the formation of coral reefs. Sancho Salcedo-Sanz[91] came up with the original idea, realizing that corals and reef formation could be artificially simulated to solve optimization problems. The TLBO method is easy to use and consumes little computational memory. TLBA is a population-based meta-heuristic optimization technique for optimizing a given objective function, and it is built on the concept of a classroom context of a school.[92] For multimodal optimization issues, FA is effective. It determines the best location for UAVs while utilizing the least energy. Based on its performance in the objective function, the algorithm assigns a light intensity to a randomly generated feasible solution known as a firefly. This intensity will be used to calculate the firefly's brightness, which is proportional to the power of its light. Another algorithm that is based on meta-heuristic memetics is the SFLA algorithm. Memetics extracts the information by evaluating cultural behavior through the lens of Darwin's evolution theory. The idea comes from frog clusters that follow a specific behavior in search of food. This unique behavior provides the foundation for the SFLA algorithm. The best frogs that can give local optimums are gathered, and the frog evolves with the wrong postures. It iterates indefinitely until an ideal path with improved convergence is found. The primary goal of this algorithm is to improve the aqueous performance of the search intensification process by employing a local search method within the framework of the genetic algorithm. Before calculating the utility of each response based on a fitness function and generating new solutions, the memetic algorithm encrypts the sum of the initial answers. It combines the advantages of particle swarm optimization and socially based genetically based memetic algorithms. The Pigeon-Inspired Optimization (PIO) algorithm is an intelligent algorithm inspired by the behavior of returning pigeons. Duan and Qiao[93] proposed it; in this algorithm, each pigeon in the swarm has a position, a velocity, and a personal best historical position, based on which it moves in the search space. An approach that can optimize the technical glitches of binary classifications is the BPIO algorithm, a binary variant of the PIO algorithm. PIO and BPIO operate by exchanging data and fostering collaboration to arrive quickly at the optimal solution.[94] The gray wolf optimizer (GWO), a population-based meta-heuristics algorithm that imitates the hierarchy of authority and hunting techniques of gray wolves in the wild, was introduced by Seyedali in 2014.[95] Gray wolves prefer to live in packs, which usually consist of 5-12 individuals. The behaviors of each wolf in pursuit of the prey are to track, chase, approach, pursue, encircle, and torment the target until it stops moving. These distinguishing characteristics of the gray-wolf provide the solution formulation for the gray-wolf optimization.

6.3 | Analysis of overall classical algorithms w.r.t area coverage

Many researchers, like Sun et al,[96] divide path planning into two major categories. The first is global path planning, also called coverage path planning, and the second is called local path planning, which includes shortest path planning. The whole environment is already known in global path planning, and the UAV plans its path on the fly according to local and surrounding conditions; this is the dynamic coverage path planning.[96] Graph research algorithms are successfully implemented for global search. These algorithms include A-star,[97] D-star,[98] and Dijkstra.[99] After graph search, random sampling algorithms are second best for global path planning. Random sampling includes probability graph algorithm[100] and rapidly exploring random trees algorithm.[101] The successful algorithms for local motion planning includes

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neural networks,[102] fuzzy logic,[103] artificial potential fields[96] and dynamic window approach.[104] In this paper, our scope is limited to local path planning in search and rescue operations. In our case, SAR is related to wilderness, and we will plan the path autonomously using 2D grids. Many scholars have suggested different approaches to performing autonomous path planning.[105] proposed a genetic algorithm for path planning in changing environments. Another approach suggested by Dhulkefl et al[106] demonstrates that dynamic programming plays a critical role in searching for a sub-optimal path in an ever-changing environment. Finally, Amin et al[107] proposed a novel approach in which they used both data structures and algorithms for optimal drone path planning. Wzorek and Doherty[108] proposed a technique for trajectory planning for autonomous UAVs. This technique used probability-based maps, and then based on these maps, rapidly exploring random trees are implemented. These techniques can optimize the path for target-oriented goals. Jevtić et al[109] initiated various intelligent solutions for drone path optimization. The results of Jevtić et al[109] show that ant colony optimization produces comparatively better results using swarm drones. Let's compare the ant colony approach with the nearest neighbor approach. It is evident that nearest neighbor is effective for local planning, and ant colony is better for global planning. Commonly, rotary wing UAVs are used in different applications. But fixed-wing drones are also used in industry. Medeiros and Da Silva[110] mentioned a technique based on the Dijkstra algorithm. In point-to-point target search, the famous Dijkstra algorithm proved very efficient. A* star algorithm is the extended form of Dijkstra by calculating all possible pathways and selecting the optimal one. The fuzzy virtual force (FVF) method was used for the path planning by Zhuoning et al.[111] The results show that FVF proved suitable for path planning as it makes this process robust and optimized. FVF was compared with one of the most used shortest path finder algorithms A*. Results show that for online path planning, FVF outperforms A*. RL is used by Kim et al.[112] They successfully implemented the technique for obstacle avoidance and safe reach to the target. They used the Q-learning version of RL.[113] controlled the UAV using radar.

6.4 | Impact of space domain

CPP is done in respect of 2D and 3D. Although in many UAV applications, altitude is fixed or remains constant, in all those cases, the approach ultimately becomes 2D, even with 3D parameters. In this section, 2D path planning is discussed. The authors in Vasquez-Gomez et al[114] proposed the algorithm for covering disjoint areas for terrain survey. They took the problem as a rural postman problem and suggested the approach to cover all the points in the area of interest. Kadioglu et al[115] suggested the idea of hexagonal tiling of the target area. In the case of agriculture-related applications with quad-rotor UAVs, they take the average distance traveled by the UAV to overcome the problem of limited battery timings. Di Franco and Buttazzo[116] proposed the technique to cover the whole searchable area with minimum energy consumption, but satisfying the coverage and time efficiency requirements. Theile et al[117] presented an approach for covering the whole area. Their method is based on DRL. They devised the data collection approach to provide the most significant coverage in the least amount of time and distance. For path planning, the researchers employed prestructure maps. Bayerlein et al[118] used a map-based approach, which is multilayered to cover all the points of interest in the target area. Optimization is based on the DDQN network. This approach creates a balance between total data collection from sensor nodes and time constraints and safety constraints.

6.5 | Issue of UAV mobility and 3D placement algorithms

In this section, we discuss the core issue due to the UAVs mobility. Especially the impacts of UAV mobility on the quality of service, packet loss ratio, coverage of the area, and resource optimization. We will also discuss how the packet loss ratio could be controlled by using 3D placement. In 3D placement, a UAV is placed at the optimal altitude to get the proper signal strength which ultimately ensures a strong link between the UAV and other communicating devices.

6.6 | Analysis of classical algorithm w.r.t known and unknown environments

Zadeh[119] used fuzzy logic to decide the path according to the current scenario. In Yang and Meng[120] and Yang et al,[105] the researchers put forward a multilayered aim-oriented path planning for dynamic environments based on fuzzy motion planning. This method has two layers; layer one decides important waypoints and is based on a global goal

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synchronized with local information. In the second layer, the drone decides an optimal path to visit all the nodes decided in layer one. This approach also avoids obstacles. Another fuzzy logic path planning algorithm was presented by Chen et al,[121] where more than one path planner was used at different stages. This approach is also used to cover the local minima and maxima problems. We use a global planner to tackle these issues. The searchable region is decomposed into smaller regions. Then a local planner is implemented for more optimality. But this approach is computationally expensive. Within a fuzzy logic domain, neuro-fuzzy is widely used in robotics. In a neuro-fuzzy system, known environments are dealt with better optimality. Previous works[122–124] all used neuro-fuzzy systems. Neuro-fuzzy logic is suitable in the case of different sensors, which are mounted on the drone. Fuzzy logic further improves the data capture and analysis by the sensors. Neuro-fuzzy system-based expert systems are also proved to be very effective. A new trend was introduced by Chen et al.[125,126] They suggested dynamic techniques for removing ice from the transmission lines. This technique was based on fuzzy logic. A known environment means having some previous data about the area. Many existing studies shows that path planning algorithms for known environments has its own impacts.[29,127,128] Famous shortest path planning algorithm Dijkstra significantly improved the best and guaranteed solution. It greatly contributed to existing path planning algorithms; this algorithm is a graph-based approach. Another excellent path optimizer algorithm was proposed by Hart et al.[29] This algorithm was famous with the name of A-star. This A-star algorithm was an extended form of the previous Dijkstra algorithm. A-star performed better in respect of the time domain using some heuristics. Later on, making A-star more robust, a multi-heuristic A-star extended algorithm was suggested for robotics path planning.[129] A-star is also able to avoid local minimum and maximum problems.

While dealing with robotics motion planning, we have to deal with many constraints like battery timings of the drones, obstacle avoidance, speed, and so forth. Many researchers focus on different rules and propose different types of solutions. For example, Latombe[130] proposed a technique for obstacle avoidance during surveying. They assumed that the environment was known. Some other researchers suggested path planners by optimizing state space. State space was minimized by changing the decomposition methods to visit the whole area by Leung et al[131] and Nilsson.[132] Later technique is based on heuristics to explore the optimal path with minimum cost from start to endpoints. For the path planning, the cost is optimized w.r.t energy consumption, distance, and time.

6.7 | Prevailing issues in overall classical algorithms

Traditional UAV path planning algorithms prioritize communication performance optimization and navigation. For instance, the goal of every graph-based algorithm like Dijkstra, A*, and D* is to identify the shortest paths. Whether the energy was optimized or not was overlooked by them. Similarly, sampling-based methods can be used in situations with unknown environments by assessing the fitness of all available samples. Although these algorithms optimize the use of resources, they are computationally expensive and inefficient in terms of time. Bio-inspired algorithms are frequently used for groups of UAVs rather than a single UAV. The fact that bio-inspired algorithms are only helpful in a limited number of situations and applications, such as locating the shared aim, is another disadvantage. The optimization of bio-inspired algorithms is further reduced when a collection of UAVs must accomplish more than one task. Any unique algorithm that cannot handle various cases is another flaw all classical algorithms share. For instance, we use different algorithms to control dynamic topology management, know and unknown environments, obstacle avoidance, and UAV swarm optimization to handle region coverage. Even if the target environment's size varies, this poses problems for a single method to solve. Manual tuning of the algorithm is required for any change or refinement. Even when using standard optimal algorithms, the resulting trajectory may not be optimal in some cases because communication performance is determined by the UAV flight trajectory and the scheduling and allocation of communication resources. Classical algorithms are required to solve communication problems, coverage, connectivity, security privacy, reduce communication energy, and overcome common path planning challenges such as path optimization, path integrity, robustness, and collision avoidance. The traveling salesman problem (TSP) is still utilized to tackle the trajectory design and communication resources in the UAV network, followed by applying several heuristic algorithms. However, this approach is only practical for trivial problems. UAV path planning using traditional techniques is often not an excellent solution to UAV communication problems in current public and industrial applications.

UAV path planning techniques based on artificial intelligence (AI) address these problems and difficulties. The growth of possible UAV applications and the advancement of drone hardware technology also brought up new challenges and related issues. Artificial intelligence and ML-based solutions to these challenges have become a hot research topic. On the other hand, ML is a catch-all term for algorithms that improve themselves automatically through

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experience and data use. Specifically, ML algorithms are classified into supervised, unsupervised or RL categories. Unsupervised algorithms infer knowledge from patterns in unlabeled data, but supervised algorithms can be guided by a labeled data set of input-output pairs. Although there is some form of feedback in RL, explicit input-output data examples are not yet available. When an agent makes sequential decisions while interacting with the environment, RL primarily builds on a trial and error problem-solving approach. RL has been particularly effective in designing UAV paths. RL-based path planning aims to find the best and shortest route and offer an automatic and learning-based solution. This solution is effective in terms of time, computational performance, energy efficiency, security, and privacy. When a deep neural network (DNN) is utilized as a part of the learning algorithm, it is called deep learning. These three ML sub-classes include deep learning adaptations. There is no universal agreement on what constitutes a "deep network" or a "shallow network," although, in general, any neural networks with more than two hidden layers are referred to as "deep networks."

The challenges associated with deploying autonomous UAVs in communication networks are particularly well suited for the RL framework. The reason is that the central idea of RL involves an independent agent making decisions to maximize some objective, such as quality of service for an aerial base station in an uncharted environment. In various cases of UAV placement and trajectory planning, such as interference-aware multi-UAV route planning, data collecting in the context of mobile crowd sensing, or increasing communications coverage in UAV-aided networks, deep RL (DRL) can be applied. Resource allocation is a UAV network application that goes beyond path planning. Without any information exchange between the UAVs, the multiagent RL (MARL) algorithm is utilized to autonomously choose each UAVs communication user, power level, and sub-channel. A user hand-off minimization technique based on RL is created in a UAV base station network. In the context of 6G, the coupling of UAVs with intelligent reflecting surfaces also produces several problems that RL can resolve. In summary, the decision-making process becomes optimal and automatic and self-learning when RL is involved in UAV path planning.

7 | ANALYSIS OF RL ALGORITHMS FOR COVERAGE PATH PLANNING

RL has important features entirely different from ML types like supervised and unsupervised learning methods. Problems requiring continuous and real-time interaction with the environment and sequential decision-making are best suited for RL. For RL, all the closed loop problems are feasible for RL. Similarly, all real-time systems where we can judge actions taken are also viable for RL. Navigational and trajectory planning systems are the best examples of RL examples. In RL, being in different situations called states, there are some available actions that can be taken. The agent's goal is to assume that action which returns maximum reward. It is deductive that the agent's duty is to map the states with activities according to some policy which maximizes the overall gain.[133] The existing algorithms for CPP with the help of RL are discussed here with their distinguishing characteristics and limitations.

7.1 | RL components

With the idea of ML, RL is very close to human learning; humans learn differently, for example, by experience. Proximity to fire teaches its hazards and usefulness. RL learns similarly. AI techniques have permeated various fields, and RL is the one which showed a significant performance increase. A survey paper of note about RL was published in 2000.[134]

In Figure 5, with feedback from its actions and experiences, RL is a machine-learning technique that enables an agent to learn in an interactive setting via trial and error. A learning agent, in general, can detect and comprehend its environment, act, and learn through trial and error. RL is efficient in sequential decision-making scenarios. RL uses a reward policy based on trial and error theory to get the most optimal solution. Figure 5 shows the basic functionality and core components of RL.

7.1.1 | Agent-environment

One of the core components of RL is the interaction between the agent and the environment. The agent makes decisions based on rewards and punishments. Consider an example of a batsman in cricket. He tries to hit the ball; if he misses, he gets a negative point. On the other hand, if he hits the ball, he gets a reward. The algorithm is the agent here,

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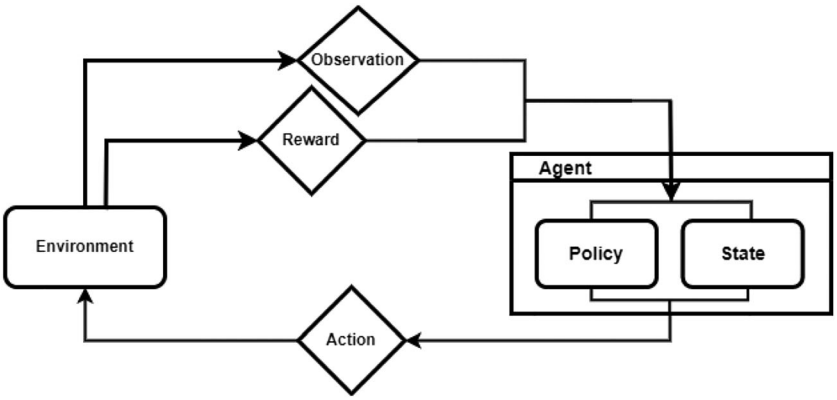

FIGURE 5 The interaction among core components in reinforcement learning

which takes action in some environments. The RL setting involves an (artificial) agent and an environment interacting with each other. The agent detects the state of the environment and takes an appropriate response. Every activity causes the environment to change to a new state governed by its dynamics (or transition model). The agent also receives from the environment a reward signal–the agent-environment interaction proceeds in several (possibly infinite) epochs. The agent's ultimate goal is determining a policy, which is a prescription (possibly stochastic) telling which action to play in every state. This form of interaction encodes a sequential decision-making problem, typically modeled with the mathematical formalism of the Markov decision processes. The goal of RL is to train an agent to complete a task within an uncertain environment.

7.1.2 | Policy

In ML, the machines learn by trial and error, wherein a reward and penalty are awarded iteratively according to some policy executed by a suitable algorithm; hence, the same operation paradigm is followed in RL. The policy is a strategy that an agent uses in pursuit of goals. The policy dictates the actions the agent takes as a function of the agent's state and the environment. Policies in RL are shrouded in a certain mystique. A policy is any function that returns a feasible action for a problem. For instance, you could take the first action that comes to mind, select an activity randomly, or run a heuristic. The better the policy, the more accurate the results will be. The policy in RL is primarily based on stimulus–response rules, which is the fundamental learning process in animals. A policy can be defined as a simple formula to determine the best option out of all possible values. This search process may become computationally expensive. Because it is discovered and develops over time via RL, the policy is regarded as being of a fundamental character.

7.1.3 | Reward

The previous section discussed policy and its foundational role in RL. Policy refinement could be done if the policy is goal oriented. If the purpose is defined, achieving that goal could serve as the metric to check for the best actions with maximum rewards. A reward is a predefined real value assigned to the agent based on its action selection. At each state, the agent selects an action out of many possible ones based on a greedy approach. But sometimes, we also make some random selections to explore other possible directions, which might lead to maximum reward ultimately. The value of the reward might range from negative to positive. An action leading to a more significant error would invoke a negative reward and vice versa. The reward calculation depends on the current state and the possible actions in that state. Being in a state, the agent takes some action based on the predefined policy; due to the action, the agent's state changes due to the transition from one state to another. Hence, the action is not only rewarded but also changes the agent's state. It is also possible that different states might have different action spaces. For example, considering a grid, we move from cell to cell. Each cell might have a maximum of four actions, that is, move left, right, up, or down. But in the case of a border cell, the agent might not have all four actions due to its boundary limits. Following is the full reward function mentioned by Sutton and Barto in their famous landmark book[135] on RL.

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$$G_t = R_{t+1} + \gamma R_{t+2} + \gamma R_{t+3} + \ldots = \sum_{k=1}^{\infty} \gamma^k R_{t+k+1}. \tag{5}$$

The reward function as described in Equation (5) is an incentive system that teaches the agent through reward and punishment, as in real life, where agents want to maximize overall profits. We must sometimes forego immediate benefits in order to maximize total rewards.

7.1.4 | Value function

For short-term benefits, selecting the best rewarding action is logical, but if we want to measure the total reward after reaching the target, the case value function is used. The benefit is the reward in the current situation; however, the value function determines the anticipated total reward if we choose the action that would result in the maximum reward. We could say that the value function tells us what action we should choose at the current state to maximize the total reward. During reward calculation, the states of every reward are also remembered. Because if an agent finds itself in a similar state next time, the agent will try to take that action with a maximum reward value. With experience, this reward function gets better, improving the results.

7.1.5 | Model

The last element of RL is the model of the environment. The model means how the environment behaves. From a present particular state and action in that state, a model can predict the next state and transition reward. The best use of the model is planning to get maximum rewards. Based on the model's predicted following states and rewards, we can measure the ultimate reward after taking some potential high-probability actions in the next states.

7.1.6 | Markov decision process

Markov decision process is fundamental to understanding the basic idea of RL. In RL, the agent takes those actions that yield maximum rewards. A point to consider is how to ensure a time-invariant reward for the same action the agent takes multiple times. It is quite possible that a reward of some arbitrary value at state 1 may have a different value when the agent takes action at state 100. How can the agent foresee the future action rewards to maximize the reward at the current state? In the following diagram, there are three primary states in the environment. Right now, the agent is at state S0. We have three options, it could stay at the current state and take no action, and the agent could go to next states. The policy of the Markov decision process decides the current state and the value of the present action in the future. In Table 7, we explained the major variants of RL with the blend of deep learning.

7.2 | Lessons learned from RL algorithms for path planning

In this section, we discuss and conclude various important points of the algorithms mentioned in Table 7. In so far discussion of the earlier sections, it is evident that path planning is more crucial in the case of disaster management.[108] Especially for UAVs, operations are also conducted in urban areas where we must consider obstacles. An intelligent and self-managed path-planning algorithm should plan the path to avoid obstacles. These aims could be achieved using traditional, and state-of-the-art AI-based algorithms.[156] Dhulkefl et al[106] described a genetic algorithm for planning the path of a drone. This approach deals with unknown and dynamic environments. Dynamic means the behavior of the environment is uncertain, it might be plain and without obstacles in one part, but it might have obstacles and nonplain terrain in another part of the searchable area.

Amin et al[107] initiated a model combining data structure and an optimization algorithm to plan the optimal path. As a data structure, it uses tree structures, and to find the optimal sample, it applies RRT algorithm, which is relatively close to the tree data structure. Wzorek and Doherty[108] proposed a technique in which they used existing path plans

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TABLE 7 Reinforcement learning for coverage path planning

Reference	RL-variant	Application	Limitations
Chen et al.[136]	Multiagent reinforcement learning	Decentralized or distributed trajectory and power control planning in wireless networks. Trajectory planning for the deployment of UAV as base station.	The model's training is centralized, but its deployment is decentralized, which raises concerns about the authenticity of the learning examples for training data. Further research is needed to determine how the overall optimality is ensured after using the decentralized strategy.
Li et al.[137]	Two-level deep reinforcement learning	Trajectory planning for the data collection in wireless sensors networks	Uncertain modeling of the environment due to partial information about the environment. Avoidance of sudden obstacles is a challenge for this technique. UAV maneuverability in 3D is also not optimal.
Cao et al.[138]	Auto-focusing heuristic trajectory planning with RL	Disaster and emergency relief operations, energy consumption efficiency	Nonoptimal for the resource utilization, User clustering and UAV scheduling in dynamic scenarios.
Benhamaid et al.[139]	Q-learning with deep learning	Data collection in wireless sensor networks with mobile nodes	Focused on cluster based trajectory planning which is not optimal for individual node cases. Trade off between energy efficiency and coverage of all the nodes is not optimal.
Apuroop et al.[60]	DRL with CNN, LSTM layer, actor-critic experience replay	Cleaning, ground surveillance robot applications	Area is decomposed into tiles for the smooth path planning but how the decomposition techniques will impact the algorithm performance is not clear. An online approach.
Theile et al.[117]	DDQN	Data harvesting and data collection in wireless network nodes	Not optimal in terms of distance traveled and energy. Specifically designed for IoT applications only.
Bayerlein et al.[118]	Double deep Q network (DDQN)	Data harvesting	An end-end reinforcement learning approach. For urban environment. Data is collected from node to node in IoT application. For optimal path planning DDQN is being used. The proposed model is based on optimized policy with balance in data collection and flight time efficiency.
Noh et al.[140]	Reinforcement learning (RL)	Cleaning robot	Especially cleaning robots or ground robots are designed for 2D motions as most of the industrial ground robotics do not deals with 3rd dimension.
Heydari et al.[141]	RL	Multipurpose UAV deployment in different scenario	Work in unknown environments, stochastic approach, finite horizon coverage, computationally not optimal to deploy in time critical operations.
Theile et al.[142]	DDQN	Flying in no-fly zones	Coverage efficiency with limited resources. Optimal calculation of the path or trajectory is not ensured.
Hong et al.[143]	DQN	Environments with obstacles	Obstacle avoidance but the optimal solution in case of sudden or moving obstacles like birds or towers is not give. Energy efficient.

(Continues)

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TABLE 7 (Continued)

Reference	RL-variant	Application	Limitations
Dong and Liu[144]	DDQN	Surveillance, monitoring for national defense and civil use	Only optimal for 2D implementation. Optimal solution to ensure 3D deployment is not clear.
Zhou et al.[145]	DRL	Multipurpose deployment in different environments.	Multirobots, works in unknown environments, obstacle avoidance, real time algorithm
Battocletti et al.[146]	RL	Unknown environments	Mapping and survey in unknown environments. Deals with critical environments. How the algorithm deals with uncertain environmental modeling is not clear. Characteristics of the algorithms are resource, time and energy efficiency.
Khalil et al.[147]	Multiagent RL	Deployment for surveillance, firefighting, cellular networks, and delivery applications	Deals with no-fly zones and obstacles. Optimality is not clear in case of sudden obstacles or moving nodes on the ground.
Devo et al.[148]	End-to-end deep reinforcement learning	Autonomous explorations	Efficient for energy, cost, weight and maneuverability
Rückin et al.[149]	DRL	Environmental monitoring and exploration	Plan the path based on the idea of coverage path or trajectory planning. Offline method. Trade off between coverage of the nodes, distance traveled and time efficiency is not clear in terms of optimal solution.
Yuan et al.[150]	Deep neural networks	Urban area UAV flight operations	Uses depth and density information contained in the images to plan the path. Might not be an optimal solution for relief or disaster management operations due to computationally high costs.
Yu et al.[151]	Deep deterministic policy gradient	General purpose	Neural net based path planning. UAV sense the environment and fit a cost function. For smooth path planning, hierarchical reinforcement learning is used. Neural nets and reinforcement learning are combined.
Yan et al.[152]	Dueling double deep Q-networks (D3QN)	Dynamic environments path planning	A deep reinforcement learning approach. Based on environmental global information. Especially designed for hostile environments. For the sake of optimization, D3QN technique is used to approximate the Q-Values.
Lim et al.[153]	Label propagation algorithm (LPA)	Shortest path finding	Multi-drone approach % Total area is divided into sub-areas for survey. % There is no specific target.
Lei et al.[154]	RL with deep learning	Coverage path planning with obstacles avoidance	Environmental mapping, path generation, CPP, and dynamic obstacle avoidance are all part of a new framework that is created in a hierarchical fashion using the environmental information that has been collected.
Hachiya et al.[155]	Q-learning	Transportation planning in disaster management operations	Preplanning is required for planning the path to supply the relief goods to the victims. Technique is still not optimal were as nature of the emergency operation demands for most optimal solutions.

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and made modifications to make them more efficient. He used the upper limits of the precomputed paths to make them more flexible toward the optimal solution. To improve the efficiency of existing techniques, the authors additionally built probabilistic maps or a rapidly explored random tree planner. The ant colony is also used to optimize the path planning algorithm.[109] The nearest neighbor method also shows promising results for selecting the best possible cases out of multiple available options. For the shortest path planning in graph theory, the Dijkstra algorithm is optimal. Medeiros and Da Silva[110] used a technique based on field height and a modified version of the Dijkstra algorithm. In the case of classical algorithms, the UAVs are consistently dependent on the ground control station during the flight. When a drone starts its flight, using the wireless network, the UAV continuously shares its current location and the collected data with the ground control station. Then the ground control station, based on the information sent by the deployed UAV, takes the other decision and instructs the UAV next step. The problem with this approach is the dependency of the field UAVs on the ground control station. If the ground control station cannot communicate with the field UAV or if there is any other distortion in wireless communication between the UAV and the ground control station, then the overall transmission will be halted. The advent of RL tackles this drawback. RL makes the UAV autonomous in its operation. First, a policy is defined to take actions according to the current environmental circumstances, then the UAV itself takes further action w.r.t path planning and connection establishment with ground users. Air-to-air communications for UAVs is the most challenging point in the context of dynamic path prediction. UAV's navigation, data routing, and communication power consumption are directly influenced by the degree of a UAV's knowledge about the context. Due to the lack of this latter, UAVs' predictions can be affected. RL solutions are trying to provide a solution for this context information by evaluating the search space and finding the optimal solution. Using global aggregated models, UAVs can forecast network resource allocation, system outages, and communication channel conditions. As a result, they can modify their trajectory, placement, and choice of the data routing path.

7.3 | Research gaps in RL algorithms

During our review of the literature on RL algorithms, we looked into UAVs' potential collaboration and cooperation and the accomplishments of various UAV roles. However, the relationship between UAVs and ground users requires more attention. These roles must be tailored to traffic dynamics and channel conditions, and both air-to-air and air-to-ground communications must be investigated collaboratively, primarily when UAVs are used. MEC. ML is an excellent solution for such a dense and dynamic environment. The most critical factors in the air to ground communication are user association and power distribution. UAVs modify their placement, trajectory, and power distribution based on user mobility and traffic distribution. A2A communication requires in-depth study to provide UAVs with information regarding users' mobility. Therefore, UAVs can modify their movement to improve the A2G connection quality and consumers' QoE. However, the prediction of user mobility and traffic distribution using UAV performance enhancement is not yet accurate and consistent.

It is evident in the discussion of RL algorithms for UAV path planning that the learning process is still nonoptimal. Policy making and learning from examples takes time which is not efficient concerning current UAV flying time and energy limitations. An example of nonoptimality is that a path is visited multiple times to reach those still unvisited areas. This approach wastes the UAVs time and energy, which already has scarce resources. Furthermore, RL is a model meant to be mature and efficient in terms of advanced features such as 3-dimensional mobility, obstacle recognition, avoidance, known and unknown environment changes, and so on. A UAV must learn to recognize when a task may be completed independently, jointly, and with cooperation and how to adjust its operation mode, computation, and offloading mechanisms in the best possible ways.

In reality, intelligence in UAVs can help them adjust their energy usage by changing the frequencies of their CPUs and the power and positioning of their transmitting antennas in response to requests. On the other hand, depending on the situation, a UAV can determine whether it is best to analyze data internally or to offload it to terrestrial MEC servers or other UAVs by understanding, for instance, the statuses of the communication links. Energy usage, delays, and user association prediction and optimization. The need for more focus on security and privacy is another issue. Assuming that UAVs are always under control, security issues have so far always been centralized. As a result, a joint autonomous defense system is required to prevent malicious network use and to ensure data integrity, confidentiality, and availability. In addition, this system must account for computing complexity, delays, and system overheads. For example, using ML to build a reputation system based on the behavior, requests, and data quality of different UAVs could be a promising direction.

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8 | CONCLUSION AND FUTURE PROSPECTIVE

This study looked at traditional and modern approaches to UAV coverage path planning using classical and RL. The capabilities and restrictions of both classes of algorithms were investigated. Different from one another, classical and RL algorithms each have their advantages and disadvantages. After the comparative analysis of classical and RL algorithms, it is evident that RL provides better autonomy. On the other hand, classical UAV path-planning algorithms are more focused on optimal solutions. This study also recommends the best mechanics for traditional path-planning algorithms and ML-based algorithms in the future. Considering the UAVs limited resources, optimization is still a crucial problem. Even the current performance of ML algorithms is adequate to demonstrate that if these algorithms are further enhanced in optimality, accuracy, autonomy, and obstacle avoidance, UAV usage will become more dependable and intensive. This study also provides future insights into the existing limitations of known and unknown environments and how the classical and RL algorithms deal with obstacles. ML's strength, when combined with UAV, is the ability to forecast context changes and optimize their performance in light of such changes. Implementing ML as a partial or worldwide solution for UAVs to carry out their tasks is almost inevitable, given the expanding number of UAVs and their flexibility and mobility. Using RL for real-world robotic tasks is challenging despite the usefulness of RL in real-world robotics and its capabilities being proved. For instance, it is not optimal for the wide exploration space and to improve sampling efficiency and safety, as well as learning that is dependable and steady and using simulations. In the long run, RL will be less effective at addressing fundamental issues such as programming robot arm movements from scratch. One of the most effective ways to integrate RL into real-world robotics more quickly, at least until research and algorithms in the field catch up, is to define problems so that RL only needs to learn what we can't model or formulate. If the problem definition in terms of actions and environment is well defined, the learning process of RL will be tremendously productive.

DATA AVAILABILITY STATEMENT

Data sharing is not applicable to this article as no new data were created or analyzed in this study.

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